HEIQ XMEMBRANE PROJECT TEAM

Final project closeout

Authoritative model-phase decision and claim boundary

2 September 2026 · Version 10

MODEL PHASE COMPLETE
TECHNOLOGY NOT VALIDATED

No existing membrane, continuous electrode mechanism, module-scale performance or net system saving has been validated. Positive energy results remain conditional model sensitivities and are not product or commercial claims.

Decision authority	Current project-closeout basis; V10 governs current decisions
Supersession	Supersedes V9 for current decisions; V9 remains unchanged as historical evidence
Selected model architecture	Direct EO transport inside the ERV membrane/core
Post-ERV O30	Coupon-test boundary/control only unless explicitly reselected later
Continuation	Separate measured-data experimental and engineering phase only
Release contents	Markdown, LaTeX, compiled PDF and SHA-256 source manifest

Executive decision

This report supersedes Version 9 for all current project decisions. Version 9 documented conditional steady-state energy windows. The later capacitive-electrode transient study materially changes their interpretation: those windows cannot be applied to long-duration operation with purely blocking electrodes. Where the reports differ, Version 10 governs.

At the original stored O30 terminal limit of 0.025 V, the primary replay retains only **16.2%** of counterfactual steady EO after one hour and **0.68%** over 24 hours. The former 98.5%/4.6% result used a separate 1.8-V current-commanded ceiling and is retained only as an optimistic upper bound.

The modelling is sufficiently mature to stop broad hypothetical parameter sweeps. It has delivered internally audited transport models, explicit architecture and comparison rules, useful sensitivity coordinates and a falsifiable coupon-test plan. It has not demonstrated a real membrane, sustainable electrode mechanism, manufacturable module or net system saving.

Final decision. The XMembrane model phase is complete. No further modelling is required to close this phase. Any continuation is a separately gated experimental and engineering phase driven by measured data.

Governing architecture

The selected modelling architecture is **direct EO transport inside the ERV membrane/core**. Local through-plane current density and water flux are evaluated along the supply and receiver channels; downstream conditioning removes any remaining load.

The historical **post-ERV O30** configuration is not a selected product or system architecture. It may be used only as a prescribed coupon-test boundary, control or benchmark, and must be labelled that way, unless a future controlled decision document explicitly selects another architecture. O30 is a humidity target coordinate, not evidence that a downstream EO module is preferred.

Interpretation rule. Historical post-ERV calculations remain provenance. They do not override the direct EO-in-ERV architecture selected for current modelling decisions.

1 Decisive new evidence

1.1 Purely blocking capacitive electrodes are transient

For two purely blocking capacitive electrodes, charge accumulates until the terminal-voltage limit is reached; current and incremental EO transport then decay toward zero. The transient solver is current-commanded with an absolute terminal-voltage clamp, unlike the voltage-controlled steady axial model. The primary replay therefore preserves each stored case's original terminal limit:

Stored case	Limit	EO 1 h	EO 24 h	Eq. time	Max pair V
O30	0.025 V	**16.2%**	**0.68%**	**0.17 h**	**0.478 V**
Full humidity duty	0.250 V	**16.5%**	**0.69%**	**0.17 h**	**0.250 V**

The separate 1.8-V current-commanded replay is an optimistic upper-bound sensitivity, not a same-voltage replay:

Stored case	Ceiling	EO 1 h	EO 24 h	Eq. time	Max pair V
O30 upper bound	1.800 V	**98.5%**	**4.6%**	**1.10 h**	**2.253 V**
Full-duty upper bound	1.800 V	**92.3%**	**5.0%**	**1.20 h**	**1.800 V**

The percentages are relative to counterfactual continuous-current EO displacement. The replay freezes the bulk states, gas films and installed area while current collapses. It is a duration diagnostic, not a dynamic, mass-balanced ERV prediction and not a revised HVAC-saving result. The reference 10,000-F/m^2 capacitance per electrode was an unmeasured sensitivity assumption.

Electrode-window limit. The terminal-voltage clamp is not an individual-electrode or electrode-pair electrochemical-window constraint. The model resolves neither individual electrode potentials nor a pair stability window. A reported pair-capacitor state may therefore exceed the terminal clamp when membrane polarisation is negative, as in the 2.253-V O30 upper bound. These are model states, not validated safe operating limits.

Symmetric, charge-balanced cycling gives **zero net EO water transport over a complete cycle** in the present constant-drag model, to numerical precision. Water moved during one half-cycle is returned during the other. A sustainable result requires a measured asymmetry or another physical mechanism; it cannot be inferred from capacitance alone.

Decision consequence. Earlier steady energy windows must not be reported as sustained performance for purely blocking electrodes.

1.2 The recent combined material case is synthetic

The recent joint $t_w = 50$, $D = 1.0 \times 10^{-10}$ m^2/s run is retained as a sensitivity result, not a candidate membrane.

Item	Result
Transport-closed rows	**4 of 8**
Best available eight-segment diagnostic area	**43,019.5 m^2**
Voltage at that diagnostic	**0.10 V**
Full-coupled/full-system rows	**0**

The audit passed integrity and reproducibility checks only. The joint property vector was not measured on an identifiable membrane; the material-validation flag is false. Area is diagnostic and the conditional energy arithmetic omits unvalidated subsystems. The run therefore does not establish a realistic material, practical module or system-energy claim.

1.3 Status of earlier energy windows

Version 9's integrated-ERV O30 result and other positive energy differences remain useful records of what the steady model predicted under its declared assumptions. They are historical conditional evidence. They must not be reported as sustained performance because continuous electrodes, same-sample membrane properties, receiver/thermal coupling and practical hardware remain unvalidated.

2 Technical reference for the frozen axial model

The steady axial calculations are **voltage controlled**: terminal voltage is held constant for a case, while current is not. Each segment solves its local through-plane current density $j(x)$, coupled to gas films and net water flux, from

$$V_{\mathrm{app}} = j(x)(\mathrm{ASR}_{\mathrm{mem}}(x) + \mathrm{ASR}_{\mathrm{series}}) + t_w \frac{RT}{F} \ln\left(\frac{a^s_{w,r}}{a^s_{w,s}}\right) + V_{\mathrm{fixed}}.$$

Activities, resistance and flux vary along the channel, so $j(x)$ varies with axial position. This is through-plane current sampled along x, not a solved in-plane current-collector current. It has equal physical magnitude on both membrane faces; its sign reverses only when written using each face's own outward normal.

Both channel gaps are 2.0 mm, so $D_h = 4.0$ mm. The model has no literal gas-film slab. At film scale factor 1, $Sh = 7.54$ gives the **derived effective film thickness**

$$\delta_{\mathrm{eff}} = \frac{D}{k} = \frac{D_h}{Sh} = 0.5305 \text{ mm}$$

on each side. At 32/20 °C, supply/receiver $k \approx 0.0511/0.0475$ m/s. The 0.5305-mm value is a correlation-equivalent boundary layer, not a physical GDL or manufactured film.

The inverse-design anchor is hypothetical: $t_w = 100.0608$, $D = 1.5979 \times 10^{-14}$ m^2/s, $\sigma = 23.1719$ S/m and dry $L = 36.9137$ μm. It must not be called a realistic membrane. The $t_w = 50$, $D = 10^{-10}$ m^2/s vector is likewise synthetic.

3 Final claim boundary

WHAT MAY AND MAY NOT BE STATED

The following statements are supported:

- the model phase is complete;
- direct EO inside the ERV is the selected modelling architecture;
- the models identify conditional sensitivities and experimental hypotheses;
- purely blocking capacitive electrodes provide finite startup displacement, not demonstrated continuous EO duty;
- the original-voltage O30 replay retains 16.2% after one hour and 0.68% over 24 hours; the 1.8-V replay is only an optimistic current-commanded upper bound;
- the $t_w = 50$, $D = 10^{-10}$ m^2/s run is a synthetic, partially closed sensitivity screen.

The following statements are **not supported** and must not be made:

- an existing or procurable membrane meets the model property bundle;
- continuous EO transport has been demonstrated;
- a Faradaic, pseudocapacitive or rectifying electrode architecture is validated;
- membrane, module, receiver, current collector or full-system performance has been experimentally validated;
- a net HVAC energy saving, commercial advantage, lifetime, cost or manufacturability target has been achieved.

Controlled closeout statement. The model phase is complete. No existing membrane, continuous electrode mechanism, module-scale performance or net system saving has been validated. All positive energy results remain conditional model sensitivities and must not be presented as product or commercial claims.

4 Accepted successor phase

MEASURED EVIDENCE BEFORE ANOTHER DESIGN CLAIM

Only the following work is accepted if the technology is continued:

1. **Demonstrate a sustainable electrode mechanism.** Measure a Faradaic, pseudocapacitive, rectifying, flow-switched or other charge/water- asymmetric mechanism. Quantify separate electrode potentials, reaction products, kinetics, overpotential, energy recovery, self-discharge and lifetime.
2. **Complete MAT-01–MAT-09 on the same identifiable sample or batch.** Measure sorption/hysteresis, mobility/permeance, EO drag, wet through-plane conductivity, membrane/contact ASR, thickness/swelling/ mechanics, coupled net water flux, interface compatibility and provenance.
3. **Run a controlled-flow coupon test.** Use matched EO-off/EO-on, ABBA polarity reversal, two-sided water balance, pressure control, film/gap repeats and complete charge/discharge cycles. Post-ERV O30 may be used only as the declared boundary/control.
4. **Close the dynamic and full-system ledgers.** Couple transient membrane inventory and gas streams, receiver operation, heat, pressure drop, fans, power electronics, heat rejection, leakage and cross-contamination.
5. **Apply product constraints.** Set accepted limits for active area, volume, pressure drop, current collection, cost, manufacturability, safety, fouling, lifetime and end-of-life handling.
6. **Replay once with measured inputs and hold a new gate review.** Do not restart broad hypothetical sweeps. Any new model or product claim requires a controlled successor report.

5 Closeout register

Workstream	Closeout status
Mathematical/model development	**Complete for this phase**
Sensitivity and architecture screening	**Complete for this phase**
Real material qualification	**Not validated; successor phase**
Sustainable electrode operation	**Not validated; successor phase**
Coupled module/full-system closure	**Not validated; successor phase**
Product/commercial decision	**No claim permitted**

Administrative closeout. The scientific modelling phase can be closed without executing the successor experimental programme. The open items are accepted future-phase gates, not incomplete deliverables of this model phase.

Release verification

The regenerated capacitive audit passes **16 of 16** checks and its targeted module passes **9 of 9** tests. The corrected targeted release suite passes **63 tests**; the full suite passes **313 tests with 1 skipped**. The sole skip is the optional design-frontier smoke test because its output is absent. These are code and integrity results, not technology validation; the governing package result remains the V10 status file.

`scripts/run_final_delivery_check_v1.py` is a superseded historical checker. It predates the September evidence and fails on stale August report and nested-manifest assumptions, so its result is not a V10 acceptance signal. The sole governing release status is the new fail-closed package verification at `deliverables/current/2026-09-02_v10/verification/V10_STATUS.json`, produced by:

```
python3 scripts/package_final_closeout_v10.py
   -expected-source-revision “$(git rev-parse HEAD)”
```

This report does not claim that package verification passes before that status file has been generated successfully.

6 Controlled evidence basis

- `reports/final_team_decision_report_2026-08-31/HeiQ_XMembrane_Final_Result_and_Decision_Report_2026-08-31_v9.tex` — preserved historical V9 report;
- `docs/CAPACITIVE_ELECTRODE_TRANSIENT_STUDY_2026-09-01.md` and `outputs/capacitive_electrode_study_v1/` — transient study and audit;
- `outputs/direct_active_erv_tw50_d1e10_v1/study/` — combined synthetic sensitivity and audit;
- `outputs/integrated_evidence_v1/material_data_requirements.csv` — MAT-01–MAT-09 qualification schema;
- `src/heiq/wp1/axial/local_state.py`, `src/heiq/wp1/airside/properties.py` and `heiq_pre_receiver_material_analyses/heiq_pre_receiver_model.py` — local voltage/current and gas-film model basis;
- `reports/final_project_closeout_2026-09-02/source_manifest.json` — SHA-256 bindings for this release and decision-critical sources.

End of controlled Version 10 closeout. V9 and earlier files remain available as historical provenance. V10 governs current project decisions.